FAULT CURRENT STUDIO - ENGINEERING REPORT
BROWSER EXPORT - ENGINEERING REVIEW REQUIRED

===== SUMMARY =====
Project: Fault Current Studio advanced study
Method: multivoltage
Model status: BOUNDED_MULTIVOLTAGE_3PH_NETWORK
Captured UTC: 2026-10-26T06:04:36.729Z
Warning: ENGINEERING SCREENING ONLY - independently verify actual manufacturer, utility, installation, equipment settings and standards edition before professional use.
Report data provenance: Entered study inputs, isolated calculated method outputs. No external OEM certification or auto-selected breaker/PPE.

===== STUDY INPUTS =====
Input path: baseMVA | Entered value: 10
Input path: frequencyHz | Entered value: 60
Input path: faultBus | Entered value: PANEL
Input path: buses[0].id | Entered value: UTILITY
Input path: buses[0].nominalLLV | Entered value: 13800
Input path: buses[1].id | Entered value: SECONDARY
Input path: buses[1].nominalLLV | Entered value: 480
Input path: buses[2].id | Entered value: PANEL
Input path: buses[2].nominalLLV | Entered value: 480
Input path: branches[0].id | Entered value: TX_1
Input path: branches[0].kind | Entered value: transformer
Input path: branches[0].from | Entered value: UTILITY
Input path: branches[0].to | Entered value: SECONDARY
Input path: branches[0].rPu | Entered value: 0.012
Input path: branches[0].xPu | Entered value: 0.055
Input path: branches[0].tapMagnitude | Entered value: 1.025
Input path: branches[0].phaseShiftDeg | Entered value: 0
Input path: branches[1].id | Entered value: FEEDER
Input path: branches[1].kind | Entered value: line
Input path: branches[1].from | Entered value: SECONDARY
Input path: branches[1].to | Entered value: PANEL
Input path: branches[1].rOhm | Entered value: 0.0012
Input path: branches[1].xOhm | Entered value: 0.003
Input path: sources[0].id | Entered value: GRID
Input path: sources[0].bus | Entered value: UTILITY
Input path: sources[0].rPu | Entered value: 0.008
Input path: sources[0].xPu | Entered value: 0.042
Input path: sources[0].emScale | Entered value: 1.1
Input path: sources[0].angleDeg | Entered value: 0
Input path: sources[1].id | Entered value: GENERATOR
Input path: sources[1].bus | Entered value: SECONDARY
Input path: sources[1].rPu | Entered value: 0.05
Input path: sources[1].xPu | Entered value: 0.19
Input path: sources[1].emScale | Entered value: 1.04
Input path: sources[1].angleDeg | Entered value: 0

===== EQUIPMENT DUTY =====
Calculated output path: status | Value: BOUNDED_MULTIVOLTAGE_3PH_NETWORK
Calculated output path: scope | Value: Fundamental-frequency balanced 3PH linear Thevenin sources + documented nominal transformer LL voltage bases, leakage on one 3PH power base and off-nominal tap phase/magnitude. No vector-group zero-sequence, non-linear fault arc, generator/motor/IBR decrement or correction factors, certified maximum/minimum IEC-60909 duty, or breaker approval.
Calculated output path: baseMVA | Value: 10
Calculated output path: frequencyHz | Value: 60

Calculated output path: faultBus | Value: PANEL
Calculated output path: faultNominalLLV | Value: 480
Calculated output path: faultMode | Value: BOLTED_FAULT
Calculated output path: faultCurrentA.real | Value: 20221.924060021163
Calculated output path: faultCurrentA.imag | Value: -59072.01134125021
Calculated output path: faultCurrentA.magnitude | Value: 62437.39854117928
Calculated output path: faultCurrentA.angleDegrees | Value: -71.10255760601106
Calculated output path: initialSymmetricalKA | Value: 62.437398541179284
Calculated output path: initialShortCircuitMVA | Value: 51.90947835155971
Calculated output path: sourceCurrents[0].id | Value: GRID
Calculated output path: sourceCurrents[0].bus | Value: UTILITY
Calculated output path: sourceCurrents[0].currentA.real | Value: 446.6515454548517
Calculated output path: sourceCurrents[0].currentA.imag | Value: -1395.2876474841898
Calculated output path: sourceCurrents[0].currentA.magnitude | Value: 1465.0342051567168
Calculated output path: sourceCurrents[0].currentA.angleDegrees | Value: -72.24938735218943
Calculated output path: sourceCurrents[1].id | Value: GENERATOR
Calculated output path: sourceCurrents[1].bus | Value: SECONDARY
Calculated output path: sourceCurrents[1].currentA.real | Value: 7059.661329898489
Calculated output path: sourceCurrents[1].currentA.imag | Value: -17954.6284794504
Calculated output path: sourceCurrents[1].currentA.magnitude | Value: 19292.680008955613
Calculated output path: sourceCurrents[1].currentA.angleDegrees | Value: -68.53552034492692
Calculated output path: branchCurrents[0].id | Value: TX_1
Calculated output path: branchCurrents[0].kind | Value: transformer
Calculated output path: branchCurrents[0].from | Value: UTILITY
Calculated output path: branchCurrents[0].to | Value: SECONDARY
Calculated output path: branchCurrents[0].fromA.real | Value: 446.651545454852
Calculated output path: branchCurrents[0].fromA.imag | Value: -1395.2876474841923
Calculated output path: branchCurrents[0].fromA.magnitude | Value: 1465.034205156719
Calculated output path: branchCurrents[0].fromA.angleDegrees | Value: -72.24938735218944
Calculated output path: branchCurrents[0].toA.real | Value: -13162.262730122671
Calculated output path: branchCurrents[0].toA.imag | Value: 41117.38286179981
Calculated output path: branchCurrents[0].toA.magnitude | Value: 43172.72673321208
Calculated output path: branchCurrents[0].toA.angleDegrees | Value: 107.75061264781056
Calculated output path: branchCurrents[1].id | Value: FEEDER
Calculated output path: branchCurrents[1].kind | Value: line
Calculated output path: branchCurrents[1].from | Value: SECONDARY
Calculated output path: branchCurrents[1].to | Value: PANEL
Calculated output path: branchCurrents[1].fromA.real | Value: 20221.924060021163
Calculated output path: branchCurrents[1].fromA.imag | Value: -59072.01134125021
Calculated output path: branchCurrents[1].fromA.magnitude | Value: 62437.39854117928
Calculated output path: branchCurrents[1].fromA.angleDegrees | Value: -71.10255760601106
Calculated output path: branchCurrents[1].toA.real | Value: -20221.924060021163
Calculated output path: branchCurrents[1].toA.imag | Value: 59072.01134125021
Calculated output path: branchCurrents[1].toA.magnitude | Value: 62437.39854117928
Calculated output path: branchCurrents[1].toA.angleDegrees | Value: 108.89744239398895
Calculated output path: busVoltages[0].bus | Value: UTILITY
Calculated output path: busVoltages[0].llV | Value: 13800
Calculated output path: busVoltages[0].phaseVoltageV.real | Value: 7580.110795780448
Calculated output path: busVoltages[0].phaseVoltageV.imag | Value: -144.6784816594609
Calculated output path: busVoltages[0].phaseVoltageV.magnitude | Value: 7581.4913796272685
Calculated output path: busVoltages[0].phaseVoltageV.angleDegrees | Value: -1.0934483963007628
Calculated output path: busVoltages[0].voltagePu.real | Value: 0.9513867409632674
Calculated output path: busVoltages[0].voltagePu.imag | Value: -0.018158730506964368
Calculated output path: busVoltages[0].voltagePu.magnitude | Value: 0.9515600193231806
Calculated output path: busVoltages[0].voltagePu.angleDegrees | Value: -1.0934483963007628
Calculated output path: busVoltages[1].bus | Value: SECONDARY
Calculated output path: busVoltages[1].llV | Value: 480
Calculated output path: busVoltages[1].phaseVoltageV.real | Value: 201.48234289577607
Calculated output path: busVoltages[1].phaseVoltageV.imag | Value: -10.220641429436764

Calculated output path: busVoltages[1].phaseVoltageV.magnitude | Value: 201.74140876379397
Calculated output path: busVoltages[1].phaseVoltageV.angleDegrees | Value: -2.9039670923628615
Calculated output path: busVoltages[1].voltagePu.real | Value: 0.7270367806739549
Calculated output path: busVoltages[1].voltagePu.imag | Value: -0.03688056300359973
Calculated output path: busVoltages[1].voltagePu.magnitude | Value: 0.7279716041029424
Calculated output path: busVoltages[1].voltagePu.angleDegrees | Value: -2.9039670923628615
Calculated output path: busVoltages[2].bus | Value: PANEL
Calculated output path: busVoltages[2].llV | Value: 480
Calculated output path: busVoltages[2].phaseVoltageV.real | Value: 0
Calculated output path: busVoltages[2].phaseVoltageV.imag | Value: 0
Calculated output path: busVoltages[2].phaseVoltageV.magnitude | Value: 0
Calculated output path: busVoltages[2].phaseVoltageV.angleDegrees | Value: 0
Calculated output path: busVoltages[2].voltagePu.real | Value: 0
Calculated output path: busVoltages[2].voltagePu.imag | Value: 0
Calculated output path: busVoltages[2].voltagePu.magnitude | Value: 0
Calculated output path: busVoltages[2].voltagePu.angleDegrees | Value: 0
Calculated output path: nodalRelativeResidual | Value: 2.761757959478921e-16
Calculated output path: assumptions[0] | Value: All complex impedances linear and at fundamental frequency
Calculated output path: assumptions[1] | Value: Transformer leakage and tap explicitly documented; leakage given on global three-phase S base
Calculated output path: assumptions[2] | Value: Transformer winding group and ground return do not enter balanced 3PH calculations
Calculated output path: assumptions[3] | Value: No fault-duty correction/decrement, CT/reclosing, or manufacturer equipment qualification

===== ASSUMPTIONS & QA =====
Evidence / limitation: Release status | Value: BOUNDED METHOD, NOT A STANDARDS/DEVICE APPROVAL
Evidence / limitation: Warning | Value: ENGINEERING SCREENING ONLY - independently verify actual manufacturer, utility, installation, equipment settings and standards edition before professional use.
Evidence / limitation: Study method | Value: multivoltage
Evidence / limitation: Timestamp | Value: 2026-10-26T06:04:36.729Z
Evidence / limitation: outputs.scope | Value: Fundamental-frequency balanced 3PH linear Thevenin sources + documented nominal transformer LL voltage bases, leakage on one 3PH power base and off-nominal tap phase/magnitude. No vector-group zero-sequence, non-linear fault arc, generator/motor/IBR decrement or correction factors, certified maximum/minimum IEC-60909 duty, or breaker approval.
Evidence / limitation: outputs.assumptions[0] | Value: All complex impedances linear and at fundamental frequency
Evidence / limitation: outputs.assumptions[1] | Value: Transformer leakage and tap explicitly documented; leakage given on global three-phase S base
Evidence / limitation: outputs.assumptions[2] | Value: Transformer winding group and ground return do not enter balanced 3PH calculations
Evidence / limitation: outputs.assumptions[3] | Value: No fault-duty correction/decrement, CT/reclosing, or manufacturer equipment qualification